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Climate change and economic growth enigma: An investment suggestion from Wall Street

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ABSTRACT

A unique opportunity to deal with climate change and stimulate faster economic growth exists today because of the 2007–2009 financial crisis coincidence with secular structural changes in the global economy. A massive investment shift from fossil to non-fossil fuels could offset sub-par economic growth and high unemployment in the developed economies caused by the financial crisis and aggravated by demographic trends and the shift of economic dynamism to emerging markets. The 2008 and dot.com crises demonstrate that our financial system is inherently unstable and that "bubbles" are not only inevitable but occur at shorter intervals and inhibit long-term investment. Financing massive investment in non-fossil fuels could be done, however, with long-term securities whose rate of return, based on exergy efficiency improvement, would attract global institutional investors with long-term investment needs. Such a massive new securities market would reduce the volatility and systemic risks inherent in today's Wall Street-with-Wall Street markets.

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1. Introduction

Wall Street has inadvertently created a unique opportunity to invest in exergy efficiency based on non-fossil resources while promoting faster economic and employment growth than can be expected today. Thanks to the financial crisis of 2007–2009 caused by the financial system ("Wall Street"), and its

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consequent Great Recession and debt crisis; combined with structural changes in the global economy and demographics, the U.S. and Europe are facing slower economic growth and higher unemployment than that experienced since WWII. Moreover, the dynamics that created the crisis of 2007–2009 are inherent in the system itself, which means that – barring fundamental change – something similar, or worse, could happen again.

To counter this grim prospect, and to mitigate the worst consequences of global climate change, wise government policy supported by financial creativity, should encourage intensive capital spending on renewable sources of energy and exergy efficiency. The large scale of investment needed to increase efficient use of non-fossil fuel and to wean the global economy from its addiction to fossil fuels could significantly boost long-term economic growth and job creation.

A major caveat: The financial community ("Wall Street") must be prevented from causing another financial crisis – especially one triggered by a resource-based asset bubble. Given the limits on the U.S.' over-extended monetary and fiscal policies today, such a bubble would be far more difficult to manage and could lead to a global depression worse than the current below-trend global economic growth.

2. The view from Wall Street

This time IS different on Wall Street. Although guilty of triggering the worst global financial crisis since 1929, which caused the Great Recession and subsequent debt crisis, Wall Street, or banks and non-bank financial institutions, were saved from bankruptcy by tax payers and the Federal Reserve. The justification for that intervention was to prevent the collapse of our financial system and another depression. In fact, only a few banks – notably Lehman Brothers – and non-bank financial institutions such as specialized insurers, were allowed to fail. This was in contrast to what happened in the 1930s, or during the Savings & Loan crisis of the 1980s (FDIC, 2013) when thousands of financial institutions went bankrupt.

Another difference: Unlike the 1930s and the S&L crisis, no senior financial figures responsible for mismanagement of their institutions and systemic risk have yet been punished, still less gone to jail. Only a few low level individuals have been prosecuted, as though they were exceptional "bad apples" and not cogs in the larger machine. Even more remarkable is that only five years after the crisis, financial markets, including both equity and debt markets, are again near all-time highs (see Fig. 1). In fact, Wall Street institutions' profits and bonuses are as exaggeratedly high as they were before the crisis erupted in 2007. Few Wall Streeters are willing to admit that their good fortune is due to their bail-out by the government, plus uniquely low interest rates (Fig. 2) and the tsunami of liquidity engineered by Quantitative Easing (QE) by the Federal Reserve since 2008 (St. Louis, 2013). They also fail to acknowledge their role in costing the middle class home-owners on "Main Street" a large fraction of their life savings, i.e. their home equity.

Another difference: After the financial shenanigans of the 1920s, strict regulatory controls such as the Glass–Steagall Banking Act of 1933 (http://www.en.wikipedia.org/wiki/Glass%E2%80%93Steagall_Act), were imposed on Wall Street. Despite the laboriously negotiated, 848-page Dodd–Frank Act of 2010 (Securities and Exchange Commission, 2013), no strict re-regulation of U.S. finance has been imposed as of summer 2013, thanks to the powerful Wall Street lobby's success in diluting and delaying the act's implementation. In reality, the financial industry seems to have been empowered by the crisis because of the "Too Big To Fail" (TBTF) phenomenon. TBTF has become the key new factor in the U.S. economy. The advocates of "universal" banking – as opposed to separation of investment banking (i.e. gambling) from conventional banking – are still in the saddle.

Knowing that if a major financial institution risks bankruptcy, the Federal Reserve and the taxpayer will be forced to bail it out (in order to preserve the national payments system) gives TBTF institutions enormous leverage. This creates a "moral hazard" that we believe is inducing Wall Street to again take on unmanageable risks. If Wall Street titans get into serious trouble, they know the Federal Reserve and taxpayers will have to bail them out yet again. And they will keep their exaggerated salaries and bonuses meaning there are almost no constraints on their willingness to take on new risk, even systemic risk.

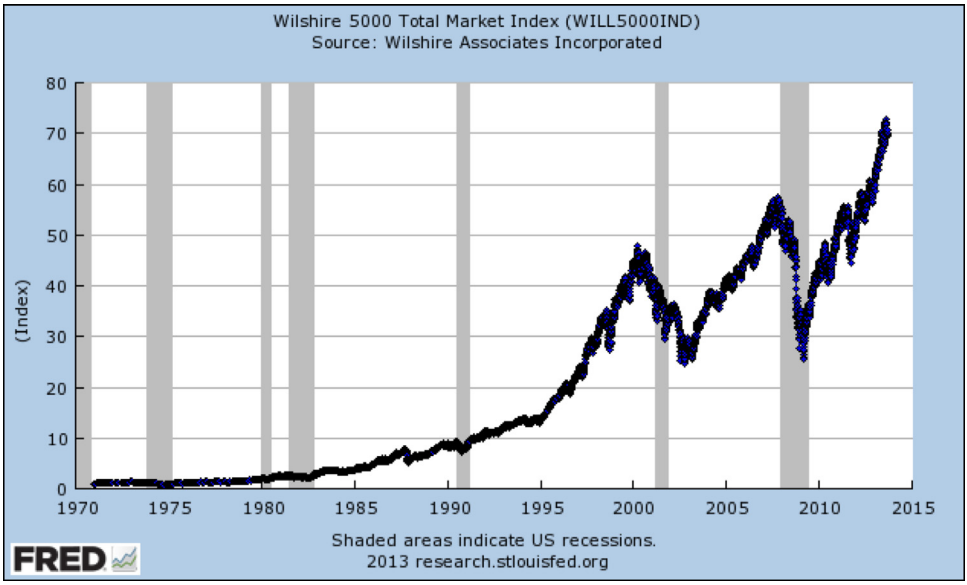

Fig. 1. U.S. equities rise with a factor 2.6 from March 2009 to new high in 2013.
Source: FRED, Federal Reserve Bank of St. Louis. The privately compiled Wilshire 5000 index (as of September 5, 2013) is the broadest index of publicly quoted U.S. companies.

This poses serious problems because U.S. policy-makers find themselves seriously constrained as a result of their huge efforts to preserve the financial system and the economy after the 2007–2009 crisis. The U.S. Congress is paralyzed because of ideological differences over the role of government and taxation, hence a fiscal policy solution is unlikely to save the system. Similarly, the Federal Reserve

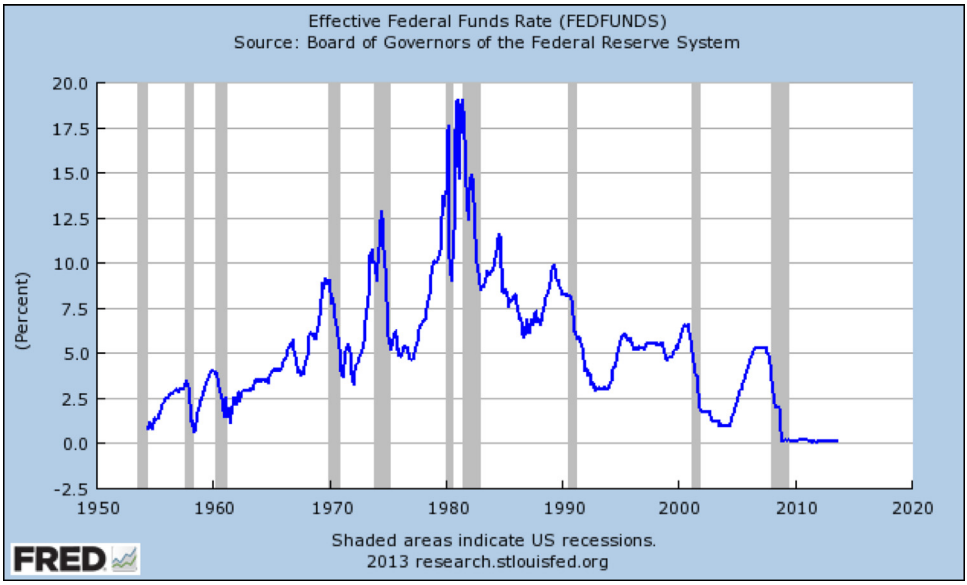

Fig. 2. Effective fed funds rate at historic low since 2007–2009 crisis.
Source: FRED, Federal Reserve Bank of St. Louis, quoting monthly data of the Federal Reserve System through August 2013.

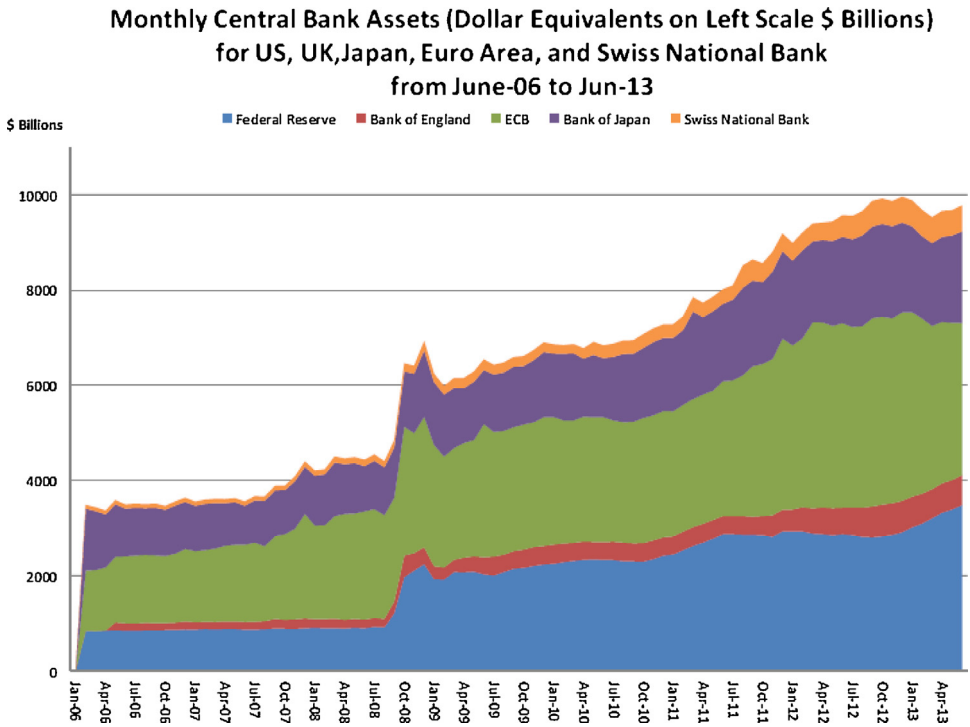

Fig. 3. G-5 central banks tripled their assets since 2007–2009 crisis. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of the article.)
Source: Cumberland Advisors, Inc. – Federal Reserve (blue), Bank of England (red), European Central Bank (green), Bank of Japan (purple) and National Bank of Switzerland (pink) from June 2006 through June 2013.

has reduced short-term interest rates to as close to zero as possible and used "Quantitative Easing" (QE, or what used to be called "printing money") to keep longer-term interest rates low. The result has been a tripling of the Fed's balance sheet (Fig. 3) and the risk that when it starts to unwind QE, financial markets may collapse.

3. Global structural differences

Additional factors make this financial crisis and its economic-social fall-out quite different from previous financial crises, the long history of which demonstrates that they are invariably followed by a depression or recession, and a long difficult recovery (Reinhart and Rogoff, 2009). This one is different, in our view, for a number of reasons. Globalization of finance since the 1990s insured that toxic securities, originated in large part by Wall Street firms to offer security through "securitization" and a few more basis points of yield, were efficiently sold into the portfolios of investors, institutional and individual, around the world. Wall Street's "anything goes" attitude was equally contagious, infecting financial institutions in Europe and Asia and exacerbating the financial crisis. The Euro zone's subsequent sovereign debt and bank crisis is the direct result of emulating Wall Street practices with a banking system that is less well capitalized and even more addicted to the "universal" model (Horne, 2013).

Aggravating the crisis' economic and social consequences is that it coincides with the doubling of the world's labor force in the last generation. The effective participation of highly motivated and low-cost Asian workers has ominous implications for the older and less motivated labor forces of Europe and North America, notably higher unemployment and less remunerative employment. Moreover,

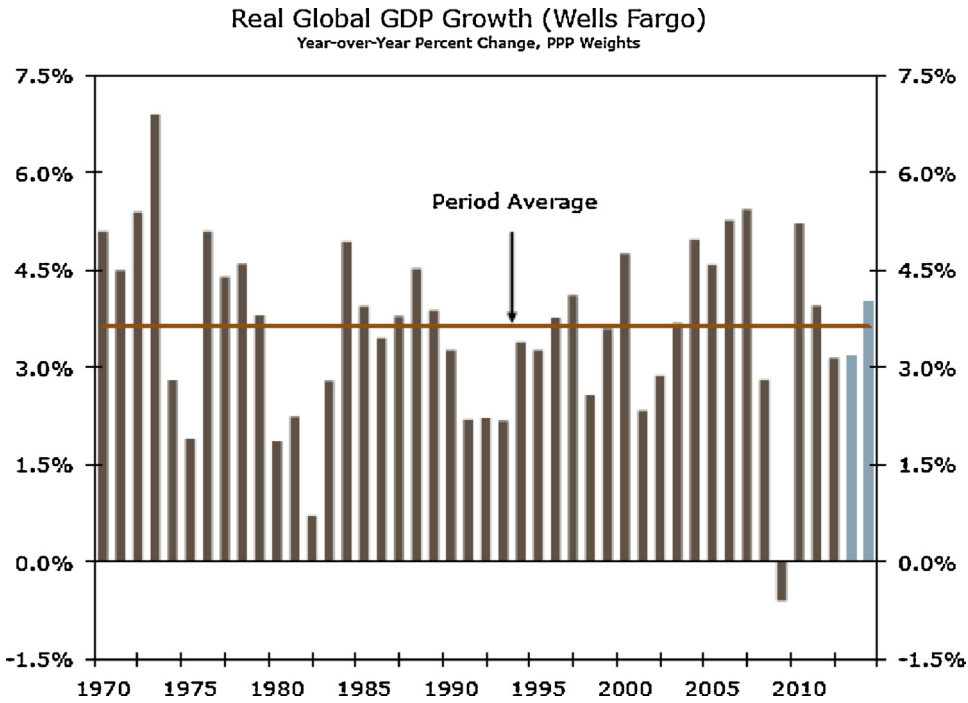

Fig. 4. Global economic growth not back to pre-crisis trend growth rate. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of the article.)
Source: Wells Fargo, "Economic Outlook", May 8, 2013. The red period average line is the average real global GDP growth rate since 1970.

substantial production capacity is being built in the emerging world, especially Asia, and is highly competitive with the aging capital stock in the U.S. and Europe.

Demographics are also an aggravating factor. Not only has the world's population more than tripled in our lifetime, to over seven billion, but it is aging rapidly in Europe, North America and even China which is aging faster than the U.S. This demographic trend means slower economic growth and a higher "dependency ratio" for the older population. Assuring a comfortable and healthy retirement for the old folks will be a huge challenge for the next generations.

This new structural global situation, plus the real risk of another financial crisis, pervasive uncertainty in the business community, the prospect of structurally higher unemployment and an aging population, all point to potential economic growth rates being substantially slower in the U.S. and Europe than we have been accustomed to since World War II. Post-crisis thinking that U.S. real GDP growth could return to 3.0%-to-3.5% per year on a sustainable basis (compared with a 2.1% average in the recovery since 2009) without generating inflation strikes us as unrealistically optimistic (Congressional Budget Office, 2013a).

With Europe stuck in a debt crisis aggravated by excessively "austerian" fiscal policies, which risk leading to an inescapable debt trap and extremist or populist political reactions; and China's growth rate decelerating; slower economic growth and higher unemployment could characterize the OECD economy for much longer than we expect. In fact, OECD and IMF medium-term forecasts point to a significantly slower rate of global economic growth, notably in the developed economies, than during the post-WWII period. (See Fig. 4.)

So, perversely, the view from Wall Street appears good thanks to the Fed-generated tsunami of low-cost capital and the TBTF "guarantee". But the view from Main Street is far less so since slower growth, higher unemployment and declining real incomes for 90% of the population mean declining living standards, growing hardship and loss of that vital American attribute: Optimism.

Clearly, we need new sources of economic growth – soon. With all evidence pointing to slowing growth in the emerging economies, plus the challenges posed by demographics and climate change, increased substitution of fossil fuels with non-fossil fuels and improved exergy efficiency are obvious candidates for accelerating investment and economic growth.

4. Policy propositions

For these reasons we agree that the five propositions made by Prof. Robert U. Ayres at INSEAD on April 10, 2013, (Ayres et al., in this volume) offer a sensible way out of our current economic impasse. I would like to add some thought to each of these. He recommends the following:

1) **The first policy proposition** is that economic policy is currently mis-focused on increasing labor productivity.

It is clear that U.S. economic growth, especially corporate profits, have been driven by increased labor productivity, meaning outsourcing jobs to lower-cost areas such as Asia; and by substitution of labor with technology-driven equipment and software. The result has been higher unemployment, reduced real compensation and skewing of the labor market toward higher skills. Most macroeconomists agree that labor productivity cannot be enhanced much more through such tactics.

2) **The second policy proposition** is that exergy waste must be actively discouraged by government, either by taxation or by regulation; and exergy efficiency must be encouraged.

Although hydraulic fracturing and other extraction technologies are producing significantly more natural gas and oil from shale deposits, causing the price of gas to decline sharply, this development weakens the incentive to reduce fossil fuel use, notably of fossil fuels with a high carbon content (in a life-cycle sense, i.e. from extraction through production to combustion). This, in turn, suggests that carbon-related climate change is likely to accelerate. Government policy through taxation of both production and use of fossil fuels, plus incentives for non-fossil fuel exergy, would help correct this negative trend. High European taxes on carbon content of fossil fuels are an example of how fossil fuel efficiency can be improved with taxation (Congressional Budget Office, 2013b). But tax restraint of fossil fuel use in the U.S. is anathema to the industry and the Republican-controlled House of Representatives.

3) **The third proposition** is that government policy needs to focus explicitly on avoiding the circumstances that lead to financial bubbles; and to compensate for income inequality and its economic-social consequences.

The apparent failure of Dodd–Frank regulatory legislation in the U.S. and the dilution and delay of Basel III (the Third Basel Accord) global, voluntary regulatory standards for bank capital adequacy, stress testing and market liquidity risk, show how important it is for the U.S. and European Union governments to act quickly on re-regulation of the financial sector in order to head of another financial crisis which could prove to be unmanageable. Transparency and accountability of TBTF institutions and trading activities are essential to increased regulation. Wall Street's first instinct is to reject transparency and accountability (Horne, 2010).

4) **The fourth proposition**: Intensive de-carbonization, to break our fossil fuel addiction.

This must be fostered by government tax policy to accelerate the long-term trend of rising real prices in the production and use of fossil fuels; as well as encouraging the attractiveness of investment in exergy based on non-fossil energy (see Congressional Budget Office, 2013a).

Most available scientific evidence points to the need to wean ourselves off of carbon-producing fuels. Given the power of the fossil fuel energy lobby, however, this will require concerted government action, supported by a public increasingly aware of the impact of global climate change. Hugely

damaging weather events, such as Hurricane Sandy and the EF5-strength tornado in Oklahoma, help boost public support for dealing with climate change.

5) **The Fifth proposition:** A practical dialog between academic experts on exergy and the long-term investment community, such as insurance companies and pension funds, to devise financial securities and investment techniques to encourage investment in non-fossil fuel-based exergy production and use.

As a market economist working with major institutional fund managers for decades, I am certain that there is a market for credible long-term securities related to exergy efficiency. But considerable work must be done to show the investment community that non-fossil fuel exergy efficiency is a viable business that can produce a sustainable and competitive rate of return for institutional investors such as pension funds and insurance companies. Government policy support will be essential to producing and marketing such a new class of securities. To be credible with the institutional investors who would buy them, however, such securities, whether fixed-income or equity, would have to be supported by market and academic data demonstrating the economic justification for improving exergy efficiency with non-fossil fuels. Academic experts on exergy such as Prof. Ayres should initiate cooperative work with insurance companies and pension asset managers, the most important of institutional investors, to provide the statistical basis needed to make new exergy-based securities credible, hence marketable.

It is clear, however, that if non-fossil fuel exergy efficiency is to become a major source of economic activity and growth, government energy, tax and environmental policies must be more forward-looking and wiser than they are today. It is equally obvious that Wall Street's financial creativity will be required to enable exergy-related financing on the very large scale that could improve our economic and climatological outlook.

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